Exponents — Laws, Domains, Conventions

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

LESSON 01 / 07

What a^x means (and when it exists)

An **exponent** a^x answers "multiply a by itself x times" only when x is a positive integer. Every other exponent is an extension built to keep the laws below true:

- $a^0 = 1$ (empty product) for every $a \neq 0$.
- $a^{-p} = \dfrac{1}{a^p}$ for $p > 0$.
- $a^{p/q} = (\sqrt[q]{a})^p = (a^{1/q})^p$ for $p \in \mathbb{Z}$, $q \in \mathbb{N}$, $q \geq 1$, $\gcd(p, q) = 1$.

Base convention (the single most important assumption of the chapter): for real exponents x that are not integers, a^x is defined for $a > 0$ only. With $a > 0$, a^x exists for every real x and is positive.

ASSUMPTIONS & CONVENTIONS

Throughout this chapter a base written a, b, c under a real exponent is assumed **positive**. If the base can be negative, only integer exponents are safe, and rational exponents need the parity rule below. Logarithm bases further satisfy $a \neq 1$. When a problem writes a^x with no comment, $a > 0$ is part of the data.

Rational exponents and parity

For $a > 0$, $a^{p/q}$ is the **unique positive** q-th root raised to p. For $a < 0$ the real q-th root exists **iff** q **is odd**. So $(-8)^{1/3} = -2$ is legal, while $(-8)^{1/2}$ is not real.

$(-8)^{2/3} = ((-8)^{1/3})^2 = (-2)^2 = 4$ — reduce p/q to lowest terms **first**, then take the odd root.

The laws of indices (and their hidden hypotheses)

For $a, b > 0$ and $r, s \in \mathbb{R}$:

$$a^r a^s = a^{r+s}, \qquad \frac{a^r}{a^s} = a^{r-s}, \qquad (a^r)^s = a^{rs},$$

$$(ab)^r = a^r b^r, \qquad \left(\frac{a}{b}\right)^r = \frac{a^r}{b^r}, \qquad a^{-r} = \frac{1}{a^r}.$$

SHORT DERIVATION

Why $(ab)^r = a^r b^r$ needs $a, b > 0$

For integer r the identity holds for all reals a, b (just expand). For $r = 1/2$ it fails on negatives: $((-1)(-1))^{1/2} = 1^{1/2} = 1$, but $(-1)^{1/2} \cdot (-1)^{1/2}$ is not real. The clean repair: restrict to $a, b > 0$, where every factor a^r is a well-defined positive number. That is exactly the convention above.

COMMON TRAP

Common trap — $(a^r)^s = a^{rs}$ is false on negatives

$((-2)^2)^{1/2} = 4^{1/2} = 2$, but $(-2)^{2 \cdot 1/2} = (-2)^1 = -2$. The rule $(a^r)^s = a^{rs}$ is valid for $a > 0$ (or when r, s are both integers). Mixing "even power" and "half power" on a negative base is the classic JEE trap. Two safe forms: $(a^2)^{1/2} = |a|$ for all real a; $a^{1/2}$ only for $a \geq 0$.

The absolute-value repairs

$$\sqrt{x^2} = |x| \text{ for all real } x; \qquad (x^{1/2})^2 = x \text{ for } x \geq 0$$

$$\sqrt[n]{x^n} = |x| \text{ if } n \text{ even}; \quad x \text{ if } n \text{ odd}$$

KEY RESULT

Three repairs worth memorising: (1) $\sqrt{x^2} = |x|$, never x; (2) reduce p/q to lowest terms before evaluating a negative base; (3) $(a^r)^s = a^{rs}$ and $(ab)^r = a^r b^r$ are stated here for $a, b > 0$ — that is the standing convention.

Solving exponent equations on the level of indices

If $a > 0$, $a \neq 1$, then $a^{f(x)} = a^{g(x)} \Leftrightarrow f(x) = g(x)$ — the exponential function is **injective** (one-to-one). If $a = 1$, both sides are 1 and the equation is empty of information; if $a = 0$ or the base can vary, handle cases.

WORKED EXAMPLE

Worked. Solve $x^{2/3} = 8$ over $\mathbb{R}$

Raise both sides to the power $3/2$ with care. Write $x^{2/3} = (x^{1/3})^2 = 8$, so $x^{1/3} = \pm 2\sqrt{2}$ and $x = \pm(2\sqrt{2})^3 = \pm 16\sqrt{2}$. Both are valid (the exponent $2/3$ has odd denominator, so $(-16\sqrt{2})^{2/3} = ((-16\sqrt{2})^{1/3})^2 = (-2\sqrt{2})^2 = 8$ ✓). Answer: $x = \pm 16\sqrt{2}$.

COMMON TRAP

Common trap — squaring and simplifying add or drop roots

$x^2 = 4 \Rightarrow x = \pm 2$, not $x = 2$. But $x^{1/2} = 2 \Rightarrow x = 4$ **only** (the left side forces $x \geq 0$ and is already non-negative; squaring here does not add roots). Before any manipulation, write the **domain** of the equation; after, check every candidate in the original equation.

LESSON 02 / 07

a^x as a function $\mathbb{R} \to (0, \infty)$

Fix $a > 0$. Then $f(x) = a^x$ has **domain** $\mathbb{R}$, **range** $(0, \infty)$, and satisfies the **exponential law**

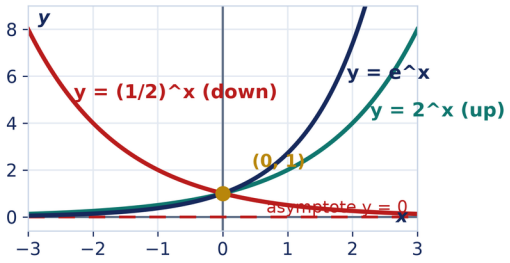
$$f(x+y) = f(x)f(y), \qquad f(0) = 1, \qquad f(-x) = \frac{1}{f(x)}.$$

(Any continuous function with $f(x+y) = f(x)f(y)$ and $f(1) = a > 0$ must be a^x — the law plus continuity pins the whole function down. Discontinuous counter-examples exist but need set-theoretic constructions beyond this syllabus; exam functions are continuous.)

- $a > 1$: strictly **increasing**, $a^x \to \infty$ as $x \to \infty$, $a^x \to 0^+$ as $x \to -\infty$.
- $0 < a < 1$: strictly **decreasing**, $a^x \to 0^+$ as $x \to \infty$, $a^x \to \infty$ as $x \to -\infty$.
- $a = 1$: the constant 1 — which is why log bases exclude 1.

Every curve $y = a^x$ passes through $(0, 1)$ and lies **above the x-axis**; the x-axis is a horizontal asymptote.

The exponential family (all pass through (0, 1), asymptote y = 0)



The number e and the natural exponential

$e = 2.718281828459045\ldots$ is the unique base for which the curve $y = e^x$ has slope 1 at $x = 0$ (the fact that will make e^x the natural exponential in calculus). Facts used constantly:

$$e \approx 2.71828, \qquad \ln e = 1, \qquad e^{x+y} = e^x e^y$$

$$(e^x)^r = e^{rx}$$

Notation: $\lg x = \log_{10} x$ (common log), $\ln x = \log_e x$ (natural log), and $a^x = e^{x \ln a}$ for every $a > 0$ — this identity converts any exponential base into e and is the workhorse of comparisons.

KEY RESULT

Monotonicity is the whole game. For $a > 1$: $a^{f(x)} > a^{g(x)} \Leftrightarrow f(x) > g(x)$. For $0 < a < 1$ the inequality **flips**: $a^{f(x)} > a^{g(x)} \Leftrightarrow f(x) < g(x)$. Same rule for log inequalities in the next sheets.

SHORT DERIVATION

Why a^x is increasing when $a > 1$ (algebraic proof)

Let $x < y$, so $y - x > 0$. Then $a^y = a^x \cdot a^{y-x}$. For $a > 1$ and $h > 0$ we have $a^h > 1$ (the q-th root of a number > 1 is > 1, and rationals are dense / continuity), so $a^y > a^x$. For $0 < a < 1$, $a^h < 1$ and the inequality reverses. This is the entire content of "exp preserves inequalities" — one multiplication by a factor > 1 or < 1.

Standard applications

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

Standard applications

1. Exponential equations by substitution. $4^x - 3\cdot 2^x + 2 = 0$: put $t = 2^x > 0$, get $t^2 - 3t + 2 = 0$, so $t \in \{1, 2\}$, i.e. $x \in \{0, 1\}$. Always keep the constraint $t > 0$ in the quadratic stage.

2. Growth comparisons. To compare 2^{300} and 3^{200}, take lg: $300\lg 2 \approx 90.31$ and $200\lg 3 \approx 95.42$, so $3^{200} > 2^{300}$. Taking a log of a positive comparison never changes its direction.

3. Mixed systems. Solve $e^{2x} - 3e^x + 2 = 0$ by $t = e^x$: $t = 1, 2$, so $x = 0$ or $x = \ln 2$.

WORKED EXAMPLE

Worked. Solve $2^{2x} - 5\cdot 2^x + 4 = 0$
$t = 2^x > 0$: $t^2 - 5t + 4 = 0 \Rightarrow t = 1$ or $t = 4$ (both > 0 ✓). So $2^x = 1 \Rightarrow x = 0$; $2^x = 4 \Rightarrow x = 2$. Check $x = 0$: $1 - 5 + 4 = 0$ ✓; $x = 2$: $16 - 20 + 4 = 0$ ✓. Answer: $x \in \{0, 2\}$.

COMMON TRAP

Common traps — exponential edition

- $(a^x)^2 = a^{2x}$ ✓ for $a > 0$; but $a^x = a^y \Rightarrow x = y$ only when $a \neq 1$.
- Substituting $t = a^x$ **throws away** the range condition $t > 0$; a quadratic giving $t = -2$ contributes nothing.
- $2^x = -4$ has **no real solution** (range of 2^x is $(0, \infty)$) — say so instead of writing $x = 2i$.
- $a^x > 0$ always: $a^x \geq 0$ or $a^x > c$ for $c < 0$ is automatically true for all x.

LESSON 03 / 07

The definition (memorise the equivalence)

For $a > 0$, $a \neq 1$ and $y > 0$:

$$\log_a y = x \iff a^x = y.$$

Read it both ways: $\log_a y$ is the **exponent** to which a must be raised to get y. Immediate consequences (all just the definition in disguise):

$$\log_a 1 = 0, \qquad \log_a a = 1, \qquad \log_a \frac{1}{a} = -1$$

$$\log_a a^p = p \quad (\forall p \in \mathbb{R})$$

$$a^{\log_a y} = y \quad (y > 0)$$

Domain discipline: $\log_a y$ requires $y > 0$ **and** $a > 0$, $a \neq 1$. $\log_a 0$ and $\log_a(\text{negative})$ are undefined in $\mathbb{R}$; $\log_1 y$ and $\log_{-2} y$ are meaningless (no unique inverse of 1^x or of $(-2)^x$).

The three laws (each comes from a law of indices)

For $M, N > 0$, $a > 0$, $a \neq 1$, $p \in \mathbb{R}$:

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^p) = p\log_a M.$$

SHORT DERIVATION

Derivation — set $M = a^m$, $N = a^n$
Then $MN = a^{m+n}$, so by definition $\log_a(MN) = m + n = \log_a M + \log_a N$. The quotient law is the same with $N \mapsto 1/N$. For the power law, $M^p = (a^m)^p = a^{pm}$ (here $M > 0$ makes $(a^m)^p = a^{mp}$ legitimate), so $\log_a(M^p) = pm$. Every log law is an index law in disguise — that is why the domain restrictions above are non-negotiable.

COMMON TRAP

Common traps — there is no law for sums or products of logs

- $\log_a(M+N)$ **does not expand**. There is no formula. Expand $\log(MN)$ or $\log(M/N)$ only.
- $\log_a M \cdot \log_a N$ is not the log of anything.
- $(\log_a M)^p \neq \log_a(M^p)$ in general: $p\log_a M = \log_a(M^p)$, but $(\log_a M)^p$ is just a power of a number.
- $\log_a(x^2) = 2\log_a x$ is **wrong for** $x<0$ — correct form: $\log_a(x^2) = 2\log_a|x|$ for $x \neq 0$.
- $\log_a a^p = p$ holds for all real p; $a^{\log_a x} = x$ holds only for $x>0$.

Change of base, chain rule, reciprocity

For $a, b, c > 0$, $a, b \neq 1$:

$$\log_a b = \frac{\log_c b}{\log_c a} = \frac{\ln b}{\ln a} = \frac{\lg b}{\lg a}$$

$$\log_a b \cdot \log_b a = 1, \qquad \log_a b \cdot \log_b c = \log_a c.$$

SHORT DERIVATION

Derivation — put $x = \log_a b$, so $a^x = b$

Take $\log_c$ of both sides: $x\log_c a = \log_c b$, hence $x = \frac{\log_c b}{\log_c a}$. Setting $c = b$ gives $\log_a b = 1/\log_b a$ (the reciprocity $\log_a b \cdot \log_b a = 1$); chaining two such fractions gives $\log_a b \cdot \log_b c = \frac{\ln b}{\ln a} \cdot \frac{\ln c}{\ln b} = \frac{\ln c}{\ln a} = \log_a c$.

The sign rule and log inequalities

For $a > 0$, $a \neq 1$, $x > 0$:

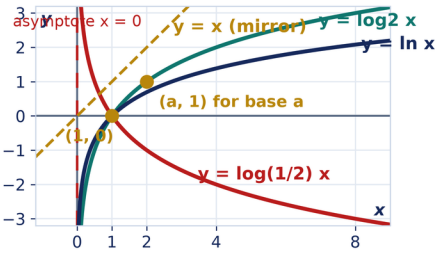
$$\log_a x > 0 \iff (x-1)(a-1) > 0$$

(equivalently: x and a lie on the same side of 1)

So $\log_2 5 > 0$, $\log_{1/2} 5 < 0$, $\log_2 0.3 < 0$.

Monotonicity: $\log_a x$ is **increasing** for $a > 1$, **decreasing** for $0 < a < 1$ — so $\log_a M > \log_a N \Rightarrow M > N$ for $a > 1$ but $\Rightarrow M < N$ for $0 < a < 1$.

Log curves (all pass (1, 0)), vertical asymptote x = 0, reflection in y = x



KEY RESULT

The graph of $y = \log_a x$ is the reflection of $y = a^x$ in the line $y = x$ (inverse functions). Both pass $(1,0)$ / $(0,1)$ respectively; log has **vertical** asymptote $x = 0$, exp has **horizontal** asymptote $y = 0$; range of log = domain of exp = $\mathbb{R}$, domain of log = range of exp = $(0, \infty)$.

WORKED EXAMPLE

Worked. Simplify $\log_2 6 - \log_2 3 + \log_2 16$

$= \log_2\left(\frac{6}{3} \cdot 16\right) = \log_2 32 = 5$. Check: $2^5 = 32 = 2 \cdot 16$ ✓.

LESSON 04 / 07

Simplification patterns that recur in exams

Nested logs peel from the inside: $\log_2(\log_3 81) = \log_2(4) = 2$. **Mixed-base sums** use change of base: $\log_4 8 = \frac{\log_2 8}{\log_2 4} = \frac{3}{2}$.

Power towers of a single base: $\log_a(a^{a^x}) = a^x$ (one application of $\log_a(a^p) = p$).

Sums of logs at fractions:

$\log_2 3 + \log_2 6 + \log_2 24 = \log_2(3 \cdot 6 \cdot 24) = \log$

.

SHORT DERIVATION

The digit formula — how many digits does N have?

A positive integer N has d digits exactly when $10^{d-1} \le N < 10^d$. Taking lg: $d - 1 \le \lg N < d$, so $d = \lfloor \lg N \rfloor + 1$. Example: 2^{100}. $\lg(2^{100}) = 100 \lg 2 \approx 30.103$, so 2^{100} has $\lfloor 30.103 \rfloor + 1 = 31$ digits ✓ (indeed $2^{100} = 1267650600228229401496703205$). The formula uses $\lfloor \cdot \rfloor$, **not rounding**, and works for any positive real N ("order of magnitude" version).

Comparing magnitudes — the three standard moves

1. **Take a log.** 2^{300} vs 3^{200}: compare $300 \lg 2 \approx 90.31$ vs $200 \lg 3 \approx 95.42$ $\Rightarrow 3^{200} > 2^{300}$. Logs are increasing, so log of two positives preserves their order (use ln or lg , any base > 1).

1. **Raise to a common power.** a^b vs b^a with $a < b$: compare after raising both to $1/(ab)$: $a^b < b^a \Leftrightarrow a^{1/a} < b^{1/b} \Leftrightarrow \frac{\ln a}{a} < \frac{\ln b}{b}$. The function $f(t) = \frac{\ln t}{t}$ increases up to $t = e$ and decreases after (calculus confirms; for the toolkit, know the shape: max at e).

1. **Squeeze between integers.** For integer estimates, bracket with known powers: $2^{10} = 1024 \approx 10^{3.01}$ is the usual bridge ($1000 \approx 2^{10}$).

WORKED EXAMPLE

Worked. Which is larger: e^π or π^e?

Take ln of both (both positive, ln increasing): $\ln(e^\pi) = \pi \approx 3.14159$, while $\ln(\pi^e) = e \ln \pi \approx 2.71828 \times 1.14473 \approx 3.112$. Since $3.14159 > 3.11229$, we get $e^\pi > \pi^e$. (Consistent with $f(t) = \frac{\ln t}{t}$ decreasing for $t > e$: here $\pi > e$, so $\frac{\ln e}{e} > \frac{\ln \pi}{\pi} \Rightarrow \pi \ln e > e \ln \pi \Rightarrow e^\pi > \pi^e$ ✓.)

Sign and size of $\log_a x$ — the grid to hold in your head

	$x > 1$	$x = 1$	$0 < x < 1$
$a > 1$	$\log_a x > 0$	0	$\log_a x < 0$
$0 < a < 1$	$\log_a x < 0$	0	$\log_a x > 0$

Absolute value: $|\log_a x| = -\log_a x$ exactly when $\log_a x \le 0$, i.e. when x and a sit on opposite sides of 1. So $\log_a |x|$ (defined for $x \neq 0$) and $|\log_a x|$ (defined for $x > 0$) are ***different functions*** — check which one a problem asks for.

KEY RESULT

Comparison checklist: (1) take ln of both sides if both are positive; (2) convert to a common base or to e via $a^x = e^{x \ln a}$; (3) for a^b vs b^a use $f(t) = \frac{\ln t}{t}$ (max at $t = e$; $2^4 = 4^2$ is the only integer pair with $a \neq b$ and $a^b = b^a$, namely $(2, 4)$ and $(4, 2)$); (4) remember $\lg 2 \approx 0.301$, $\lg 3 \approx 0.477$, $\ln 2 \approx 0.693$ — three numbers that unlock most numerical estimates.

COMMON TRAP

Common traps — comparison edition

- Taking log of an inequality requires ***both sides positive***; log of a negative is undefined, not "negative log".
- $\lg(a + b) \neq \lg a + \lg b$; write $\lg(a + b)$ as it is or factor first.
- $\log_a x$ increasing is for $a > 1$ only; with $0 < a < 1$, $\log_a x > \log_a y \Rightarrow x < y$.
- $2^{300} \approx 10^{90.3}$ means 2^{300} is between 10^{90} and 10^{91} — use floors, not rounding, for digits.

LESSON 05 / 07

The four structural types

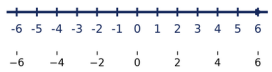
Type 1 — common base. $a^{f(x)} = a^{g(x)}$ with $a > 0$, $a \neq 1 \Rightarrow f(x) = g(x)$. If the bases differ but are related ($4^x = 8^{x-1}$), convert: $2^{2x} = 2^{3x-3} \Rightarrow x = 3$.

Type 2 — quadratic in an exponential. $a^{2x} + Ba^x + C = 0$: substitute $t = a^x > 0$, solve, then reject $t \leq 0$ and take $x = \log_a t$.

Type 3 — logarithmic equation to exponential. Isolate a single log, then rewrite: $\log_2(x-3) = 4 \Rightarrow x - 3 = 2^4 \Rightarrow x = 19$. If logs remain, use laws to **collapse** to one log first: $\log_2(x-3) + \log_2(x+1) = 3 \Rightarrow \log_2((x-3)(x$.

Type 4 — x in base and argument. $x^{\log_2 x} = 16$-type: take $\log_2$: $(\log_2 x)^2 = 4 \Rightarrow \log_2 x = \pm 2 \Rightarrow x = 4$ or $x = \frac{1}{4}$. Both check out (a power x^y with $x > 0$ is always legal).

Domain checks turn 'solutions' into verified roots — worked example 2: only x = 2 + 2√3 survives



The domain-first discipline

Every log equation has an **implicit domain**: each $\log(\cdot)$ needs its argument > 0; every a^x is fine on $\mathbb{R}$ for $a > 0$; $x^{p/q}$ needs the parity care of sheet 1. Solve **first** on the domain, then verify — or solve freely and **filter at the end**. Either way the final answer must be checked in the original equation, because squaring, exponentiating, and multiplying by logs can create ghosts.

SHORT DERIVATION

Worked 1. $2^{2x} - 5 \cdot 2^x + 4 = 0$ **(type 2)**
$t = 2^x$, $t > 0$: $t^2 - 5t + 4 = 0 \Rightarrow t = 1, 4$, both admissible. $2^x = 1 \Rightarrow x = 0$; $2^x = 4 \Rightarrow x = 2$. Substitute back: $x = 0$ gives $1 - 5 + 4 = 0$ ✓; $x = 2$ gives $16 - 20 + 4 = 0$ ✓. Answer $\{0, 2\}$.

SHORT DERIVATION

Worked 2. $\log_2(x-3) + \log_2(x+1) = 3$ **(type 3)**
Domain: $x > 3$ **and** $x > -1$, so $x > 3$. Collapse: $\log_2((x-3)(x+1)) = 3 \Rightarrow (x-3)(x+1) = 8$. Only $1 + 2\sqrt{3} \approx 4.464$ lies in $x > 3$; $1 - 2\sqrt{3} < 0$ is a ghost created by the algebra (it would make the original logs undefined). Answer: $x = 1 + 2\sqrt{3}$.

WORKED EXAMPLE

Worked 3. $x^{\lg x} = 10^4$ **(type 4)**
Take $\lg$ of both sides (both positive): $(\lg x)(\lg x) = 4 \Rightarrow (\lg x)^2 = 4 \Rightarrow \lg x = \pm 2 \Rightarrow x =$ or $x = 10^{-2} = 0.01$. Check: $100^{\lg 100} = 100^2 = 10^4$ ✓; $(0.01)^{\lg 0.01} = (10^{-2})^{-2} = 10^4$ ✓. Both answers valid.

COMMON TRAP

Common traps — equations edition

- **Collapsing logs** $\log A + \log B = \log(AB)$ requires $A, B > 0$ — the domain shrinks, it does not grow.
- Squaring both sides (e.g. when clearing $\sqrt{\cdot}$ or $|\cdot|$) creates ghost roots — verify each one.
- $\log_a f = \log_a g \Rightarrow f = g$ **plus** $f, g > 0$; never cancel logs before writing the domain.
- Exponentiating $e^A = e^B$ is reversible for all real A, B; but $A^2 = B^2 \Rightarrow A = \pm B$ — do not "cancel squares".
- If a base depends on x ($x^{\log x}$, $\log_x 2$), add the conditions $x > 0$, $x \neq 1$ (base rule) on top of argument rules.

LESSON 06 / 07

Log and exponent inequalities

Solve like equations, then respect the monotonicity direction. Worked: $\lg(x-1)<1$. Domain $x>1$; lg is increasing ($10>1$), so $x-1<10^1=10$; combine: $1<x<11$.

With a base <1 the inequality flips: $\log_{1/2}(x-1)<1 \Rightarrow x-1>(1/2)^1=\frac{1}{2}$ (flip!) $\Rightarrow x>\frac{3}{2}$.

Exponential inequalities follow the same rule: $2^{x^2-1}\geq 2^{3x} \Rightarrow x^2-1\geq 3x$ (base $2>1$) $\Rightarrow x^2-3x-1\geq 0 \Rightarrow x\leq \frac{3-\sqrt{13}}{2}$ or $x\geq \frac{3+\sqrt{13}}{2}$.

KEY RESULT

Inequality protocol: (1) write the domain first (logs: argument >0); (2) bring to one monotone function and read off the argument inequality (flip if base <1); (3) intersect with the domain; (4) for $|\log|$ or mixed signs, split cases before step 2. Parametric problems: solve for x with the parameter as a constant, then discuss the parameter ranges where the solution set changes (usually at the boundary points where a log argument hits 0/1 or a base equals 1).

Number-of-digits and order-of-magnitude models

$d(N)=\lfloor \lg N\rfloor+1$ (see sheet 4). Variants: how many integers $n\leq N$ satisfy something? take logs; how many powers 2^k lie in $[a,b]$? count k with $\lg a\leq k\lg 2<\lg b$.

WORKED EXAMPLE

Worked. How many digits does 3^{50} have? First two digits estimate.
$\lg(3^{50})=50\lg 3\approx 23.856$. So $d=\lfloor 23.856\rfloor+1=24$ digits. For the leading digits: $10^{0.856}\approx 7.18$, so $3^{50}\approx 7.18\times 10^{23}$ — leading digit 7.
(Check: $3^{50}=717897987691852588770249$ ✓ 24 digits, starts with 7 ✓.)

Growth models (the syllabus-fit applied framing)

Populations, compound interest, and radioactive decay all use $N(t)=N_0a^t$ or N_0e^{kt}; with a *fixed* percentage growth per period the multiplier is constant and the index is time. Doubling time for e^{kt}: $kt=\ln 2$, so $t=\ln 2/k$ (with $\ln 2\approx 0.693$). Half-life is the same with $-\ln 2$. These are **models built from the supplied function** a^x — the exam task is always to translate words into a^x or log, then solve with the toolkit above.

Comparison of growth classes (use as ordering, not as calculus): as $x\to\infty$ (any fixed bases $a>1$, $b>1$, $c>1$): $\log_c x \ll x^{1/n}$ (for every n) $\ll a^x$. "Slowest to fastest: log, then root/power, then exponential." This is why compound growth overtakes polynomial growth eventually, and why taking logs linearises exponential comparisons.

SHORT DERIVATION

Solving $a^x=b^{x+c}$ — the canonical parameter problem
Take ln:

$$x\ln a=(x+c)\ln b \Rightarrow x(\ln a-\ln b)=c\ln b \Rightarrow x$$

, provided $a\neq b$ (if $a=b$: either no solution ($c\neq 0$) or all x ($c=0$)). Every "two-exponential crossing" problem is this shape after factoring.

COMMON TRAP

Common traps — applications edition

- Word problems smuggle in domains: "log of the population" assumes the population is positive; "after t years" may mean $t \in \mathbb{N}$ vs $t \geq 0$ real.
- Doubling/half-life formulas use **natural** logs; mixing $\lg$ and $\ln$ in one computation is a classic arithmetic error.
- $\lfloor \lg N \rfloor + 1$ counts digits of the **integer part** of N; for non-integers state the convention used.
- Growth "per cent per year" compounds: 1.06^t, not $1 + 0.06t$ (that linear model is a different, weaker statement).

LESSON 07 / 07

The whole toolkit on one sheet

Exponents ($a > 0$ convention). $a^r a^s = a^{r+s}$; $(a^r)^s = a^{rs}$ (needs $a > 0$); $(ab)^r = a^r b^r$ (needs $a, b > 0$); $a^{-r} = 1/a^r$; $a^0 = 1$ (empty product, incl. $0^0 = 1$ as convention). Rational p/q in lowest terms; negative base only with odd q. $\sqrt{x^2} = |x|$.

Exponential function. Domain $\mathbb{R}$, range $(0, \infty)$, through $(0, 1)$, asymptote $y = 0$; $a > 1$ increasing, $0 < a < 1$ decreasing; $f(x + y) = f(x)f(y)$; $a^x = e^{x \ln a}$.

Logarithm. $\log_a y = x \Leftrightarrow a^x = y$; domain $y > 0$, base $a > 0$, $a \neq 1$; laws $\log MN = \log M + \log N$, $\log(M/N) = \log M - \log N$, $\log M^p = p \log M$ (with $M > 0$; repair $\log x^2 = 2\log|x|$); change of base $\log_a b = \frac{\ln b}{\ln a}$; chain $\log_a b \cdot \log_b c = \log_a c$; inverses $a^{\log_a x} = x$ ($x > 0$), $\log_a(a^x) = x$.

Signs and monotonicity. $\log_a x > 0 \Leftrightarrow (x - 1)(a - 1) > 0$; $\log_a$ increasing for $a > 1$, decreasing for $0 < a < 1$ (flip inequalities); $\lg 2 \approx 0.301$, $\lg 3 \approx 0.477$, $\ln 2 \approx 0.693$, $e \approx 2.718$.

Equations. Types: common base ($f = g$), quadratic in a^x ($t = a^x > 0$), collapse-logs-then-exponentiate (domain first!), $x^{\log x}$ (take logs). Always: domain, solve, verify in the original.

Applications. Digits $d = \lfloor \lg N \rfloor + 1$; compare magnitudes by $\ln$; a^b vs b^a via $\frac{\ln t}{t}$ (max at e); growth $N_0 a^t$ / $N_0 e^{kt}$, doubling $t = \ln 2/k$; growth hierarchy log $\ll$ power $\ll$ exponential.

KEY RESULT

Five questions to ask any exponent/log problem. (1) What is the domain (bases > 0, arguments > 0, bases $\neq 1$)? (2) Which single monotone function can I bring everything under? (3) Is the base > 1 or < 1 (inequality direction)? (4) After solving, which candidates survive the original equation? (5) Are units/bases mixed ($\lg$ vs $\ln$, degrees vs real exponents)?

ASSUMPTIONS & CONVENTIONS

Conventions fixed for the whole chapter: a^x real-valued for $a > 0$ (negative bases only through integer exponents or odd roots); $0^0 = 1$ as empty-product convention; $\lg = \log_{10}$, $\ln = \log_e$; all logs real, so arguments positive; "log" with no base means a formula holds for every legal base.

STANDARD APPLICATION

Where this toolkit reappears. Domain/monotonicity discipline powers calculus limits (1^∞ forms via $\ln$), analysis of $f(t) = \frac{\ln t}{t}$, and counting estimates (digits, digits of $n!$, order of magnitude) in later chapters and in Olympiad number theory (Legendre's formula — see Part II). The habit "convert a^x to $e^{x \ln a}$" is the bridge to differential calculus.

COMMON TRAP

Final trap sweep

- $\log_a(M+N)$ has **no** expansion; $(\log M)^p \neq \log M^p$.
- $(a^r)^s = a^{rs}$ and $(ab)^r = a^r b^r$ silently need $a, b > 0$.
- Inequality flips when the base is in $(0, 1)$.
- Ghost roots from squaring / collapsing logs: verify in the original.
- $2^x = c$ has a real solution only for $c > 0$; then $x = \log_2 c$ uniquely.

LESSON 01 / 02

Status of this sheet

The equation $x^y = y^x$ and multi-base logarithmic systems are **standard IOQM/RMO algebra** and appear in specialised exam questions; everything below is derived from the Part I toolkit (monotonicity, change of base, $\frac{\ln t}{t}$) with no new theory assumed. Nothing here "can never appear in JEE" — the classification is only that these models go beyond the current syllabus sentence "Logarithms and their properties".

$x^y = y^x$ — the complete positive solution set

For $x, y > 0$: if $x = y$ everything works. If $x \neq y$, take ln: $y\ln x = x\ln y$, i.e. $\frac{\ln x}{x} = \frac{\ln y}{y}$. So x, y are two distinct points where $f(t) = \frac{\ln t}{t}$ takes the **same value** — and f increases on $(1, e]$, decreases on $[e, \infty)$, so any level is cut at most twice.

SHORT DERIVATION

Parametrisation of all unequal solutions

Put $y = tx$ with $t > 0$, $t \neq 1$. Then $x^{tx} = (tx)^x \Rightarrow x^t = tx \Rightarrow x^{t-1} = t \Rightarrow x = t^{1/(t-1)}$, $y = t^{t/(t-1)}$. Writing $t = 1 + \frac{1}{n}$ gives the classic family $x = \left(1 + \frac{1}{n}\right)^n$, $y = \left(1 + \frac{1}{n}\right)^{n+1}$ for $n > 0$ ($n \in \mathbb{Q}^+$ or $\mathbb{N}$).

Check $n = 1$: $(x, y) = (2, 4)$: $2^4 = 4^2 = 16$ ✓. Integer solutions with $x \neq y$: exactly $(2, 4)$ and $(4, 2)$.

WORKED EXAMPLE

Worked. Real solutions with $x < y$, $xy = 8$, $x^y = y^x$?

$\frac{\ln x}{x} = \frac{\ln y}{y}$ with $y = 8/x$; solve $\frac{\ln x}{x} = \frac{\ln(8/x)}{8/x}$ numerically/graphically by monotonicity arguments; or use the parametrisation: $\frac{y}{x} = t = 1 + \frac{1}{n}$, $xy = 8$ fixes n. From $y = tx$ and $xy = 8$: $tx^2 = 8$. With $x = t^{1/(t-1)}$ this becomes $t^{1 + \frac{2}{t-1}} = 8 \Rightarrow t^{\frac{t+1}{t-1}} = 8$. Trial $t = 4$: $4^{5/3} \approx 10.08 \neq 8$; $t = 2$: $2^3 = 8$ ✓. So $t = 2$, $x = 2$, $y = 4$. Answer: $(x, y) = (2, 4)$ (and the symmetric pair if order is free).

Multi-base logarithmic systems

The three moves: (1) reciprocity $\log_a b \cdot \log_b a = 1$; (2) chain $\log_a b \cdot \log_b c = \log_a c$; (3) change every base to ln (or to a common base c) and clear denominators.

WORKED EXAMPLE

Worked. Solve $\log_2 x = \log_x 4$

Domain $x > 0$, $x \neq 1$. Reciprocity: $\log_x 4 = \frac{\ln 4}{\ln x} = \frac{2\ln 2}{\ln x}$; also $\log_2 x = \frac{\ln x}{\ln 2}$. Set $u = \ln x$: $\frac{u}{\ln 2} = \frac{2\ln 2}{u} \Rightarrow u^2 = 2(\ln 2)^2 \Rightarrow u = \pm\ln 2 \cdot \sqrt{2}$. So $x = 2^{\sqrt{2}}$ or $x = 2^{-\sqrt{2}}$. Check both are > 0 and $\neq 1$ ✓. Answer: $x \in \{2^{\sqrt{2}}, 2^{-\sqrt{2}}\}$.

Towers obey a convention: $a^{b^c} = a^{(b^c)}$ — exponentiation is **right-associative**; $(a^b)^c = a^{bc} \neq a^{b^c}$ in general. Nested logs peel from the inside: $\log_2(\log_3(\log_4 256)) = \log_2(\log_3 4) = \log_2\left(\frac{\ln 4}{\ln 3}\right)$ — leave as exact expressions; never invent a "log of log" law.

COMMON TRAP

Traps on this sheet

- $x^y = y^x$ over **integers** vs **reals** give different answer sets; the parametrisation covers reals; integers need $t = 1 + 1/n$ with t rational $\Rightarrow n = 1$ only among integers (for $x < y$ positive integers: $\{2, 4\}$).
- $\log_a b = \log_a c \Rightarrow b = c$ **needs the same base**; $\log_2 x = \log_3 x \Rightarrow x = 1$ only.
- Solving $\log_x(\cdot)$ adds base conditions $x > 0$, $x \neq 1$ on top of argument positivity.
- $f(t) = \frac{\ln t}{t}$ is **not** monotone on $(0, \infty)$ — it turns at $t = e$; two solutions for $f(t) = c \in (0, 1/e)$ sit on opposite sides of e.

Assumptions & conventions

All variables positive reals unless stated; integer solutions are a separate reading of the same equations; $\frac{\ln t}{t}$ analysis uses its stated monotonicity shape (derivable later by calculus; here taken as the known shape of the standard curve, with the level-curve picture as the geometric model).

LESSON 02 / 02

Status of this sheet

Order-of-magnitude estimates and Legendre's formula are **classic IOQM/RMO number theory** built from the Part I toolkit (logs, floors, geometric progressions). The hierarchy statements below are comparisons between standard functions, models derivable from supplied information — not additional theory claimed as a syllabus item.

The growth hierarchy (state it as ordering, use it as a tool)

For fixed $a > 1$, $b > 1$, $c > 1$, $k \in \mathbb{N}$, as $x \to \infty$:

$$\log_c x \ll x \ll x^k \ll a^x \ll b^{b^x}.$$

Use in estimates: to compare $n^{\log n}$-shaped quantities, take $\log$ once or twice until everything is polynomial. Example: which is larger for huge n — 100^n or n^{100}? Take ln: $n \ln 100$ vs $100 \ln n$; divide by n: $\ln 100 \approx 4.605$ vs $\frac{100 \ln n}{n} \to 0$. So 100^n wins for large n; at $n = 100$ the two are **equal**, for $n > 100$ the exponential 100^n is larger, for $e < n < 100$ the power n^{100} is larger (crossover where $\frac{\ln n}{n} = \frac{\ln 100}{100} = 0.046$).

KEY RESULT

Digit-counting engine. $d(N) = \lfloor \lg N \rfloor + 1$ for integers $N \geq 1$. Sums of digits in base b: $N = \sum d_i b^i$ with $0 \leq d_i < b$; the digit-sum $s_b(N) = \sum d_i$ satisfies $N - s_b(N)$ is divisible by $b - 1$ (casting out $b - 1$'s). This pair ($\lfloor \log \rfloor$, digit sum) is exactly what Legendre's formula packages.

Legendre's formula — the exponent of a prime in $n!$

For prime p: $v_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor = \frac{n - s_p(n)}{p - 1}$,

where $s_p(n)$ is the digit sum of n in base p (finitely many non-zero terms: stop when $p^k > n$).

SHORT DERIVATION

Derivation — count multiples, then clean up with digits

Among $1, 2, \ldots, n$, exactly $\lfloor n/p \rfloor$ are divisible by p; among those, $\lfloor n/p^2 \rfloor$ contribute a **second** factor p; and so on — each integer m contributes $v_p(m)$ factors, so the total is $\sum_{k \geq 1} \lfloor n/p^k \rfloor$ ✓ (count each m once per power of p dividing it). For the closed form, write $n = s_p(n) + (p - 1) \sum_{k \geq 1} \lfloor n/p^k \rfloor$ (base-p identity: $\lfloor n/p^k \rfloor = \sum_{j > k} d_j p^{j-k-1}$), rearranged to the formula.

WORKED EXAMPLE

Worked. $v_2(100!)$ and $v_5(100!)$
$v_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97$. *Via digits:* $100 = (1100100)_2$, $s_2 = 3$, $\frac{100-3}{1} = 97$ ✓. $v_5(100!) = 20 + 4 = 24$; *digits:* $100 = (400)_5$, $s_5 = 4$, $\frac{100-4}{4} = 24$ ✓. *Consequence:* $100!$ *ends in exactly* 24 *zeros (the power of* $10 = 2 \cdot 5$ *is the smaller of* v_2, v_5*).*

Series estimates with logs and floors

Truncation errors: $\sum_{k=1}^{n} \frac{1}{k}$ *grows like* $\ln n$ *(harmonic estimate:* $\ln(n+1) \le H_n \le 1 + \ln n$ *— provable by comparing the sum with* $\int \frac{dx}{x}$*, a standard derived model). The telescoping estimate* $\frac{1}{k+1} \le \ln\left(1 + \frac{1}{k}\right) \le \frac{1}{k}$ *(from* $\frac{1}{1+t} \le \ln(1+t)$*-style log bounds at* $t = 1/k$*) is the workhorse behind it.*

WORKED EXAMPLE

Worked. Bound $H_{100} = 1 + \frac{1}{2} + \cdots + \frac{1}{100}$
$\ln 101 \approx 4.615 \le H_{100} \le 1 + \ln 100 \approx 5.605$ *; the true value is* ≈ 5.187 ✓ *inside the bracket. Use such brackets to decide sign/size questions without computing sums.*

COMMON TRAP

Traps on this sheet

- $\lfloor n/p \rfloor + \lfloor n/p^2 \rfloor + \cdots$ *is* ***not*** *a geometric series — do not sum it as one; each term is a floor.*
- *Trailing zeros of* $n!$ *are* $v_5(n!)$ *(5s are rarer than 2s), not* $v_2(n!)$*.*
- log *estimates compare growth rates; a bound valid "for large* n*" must not be quoted for a specific small* n *without checking (e.g.* $n^{100} > 100^n$ *fails for* $n = 2$*).*
- *Digit sums* s_p *depend on the base;* $s_2(100) = 3$ *counts* ***ones in binary****, not decimal digits.*

Assumptions & conventions

p prime; $v_p(m)$ = exact exponent of p in m; floors are real floors; harmonic estimates quoted as standard derived bounds (checked numerically above); hierarchy statements are eventual comparisons (there is always a crossover point).